

Hazardous waste site listed under the
Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (CERCLA "Superfund")

KYSOR INDUSTRIAL CORP.
Cadillac, Michigan

Kysor Industrial Corp. manufactures temperature control systems for the automotive industry on a 0.10-acre site in Cadillac, Wexford County, Michigan. The process involves stamping and machining metal parts. Prior to 1979, 665 cubic yards of liquid/sludge wastes containing solvents (1,1,1-trichloroethane, trichloroethylene, toluene, and ethyl benzene) used to clean metal parts were disposed of in unlined earthen pits on the site. In 1981, the company excavated the pits and sent the materials to an off-site disposal facility.

On-site monitoring wells that tap shallow ground water are contaminated with solvents, including toluene and trichloroethylene, according to tests conducted by consultants to the company.

A shallow aquifer within 3 miles of the site provides water for 4,500 people, approximately 8 percent of Cadillac's population. The nearest surface water (0.4 miles downstream from the facility) is used for fishing.

A container storage area at Kysor received Interim Status under the Resource Conservation and Recovery Act when the company filed Part A of a permit application. In July 1984, EPA approved a plan for closing the area.

EPA Region 5 Records Ctr.



234876

Facility name: <u>Kysor INDUSTRIAL CORP.</u>	
Location: <u>1100 Wright St. Cadillac, MI 49601 Wexford Co.</u>	
EPA Region: <u>05</u>	
Person(s) in charge of the facility: <u>David Crooks</u> <u>Mike Mickelson</u>	
<u>Vice Pres/General Counsel</u>	<u>Plant Engineer</u>
<u>(616) 775-4646</u>	<u>(616) 775-4681</u>
Name of Reviewer: <u>Ronald Bock</u>	Date: <u>4/11/85</u>
General description of the facility: (For example: landfill, surface impoundment, pile, container; types of hazardous substances; location of the facility; contamination route of major concern; types of information needed for rating; agency action, etc.)	
<u>Kysor manufactures automotive and truck engine temperature control systems by stamping and machining metals. Some of the finished parts are painted. Prior to 1979, wastes were disposed in earthen pits on site. On July 22-23, 1981, pits were excavated and the contaminated materials were sent to Wayne Disposal. Presently, wastes are stored in drums on site until transport. Monitoring wells at Kysor indicate high levels of contamination which could contaminate private wells. In fact, there has already been a documented case of a private well being contaminated which is likely linked to Kysor.</u>	
Scores: $S_M = 33.94$ $(S_{gw} = 55.71)$ $S_{SW} = 0$ $S_a = 0$	
$S_{FE} = 0$	
$S_{DC} = 0$	

FIGURE 1
HRS COVER SHEET

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Ground Water Route Work Sheet						
Rating Factor	Assigned Value (Circle One)	Multi-plier	Score	Max. Score	Rel. (Section)	
1 Observed Release	0 (45)	1	45	45	3.1	
If observed release is given a score of 45, proceed to line 4. If observed release is given a score of 0, proceed to line 2.						
2 Route Characteristics					3.2	
Depth to Aquifer of Concern	0 1 2 3	2		6		
Net Precipitation	0 1 2 3	1		3		
Permeability of the Unsaturated Zone	0 1 2 3	1		3		
Physical State	0 1 2 3	1		3		
Total Route Characteristics Score			NA	15		
3 Containment	0 1 2 3	1	NA	3	3.3	
4 Waste Characteristics					3.4	
Toxicity/Persistence	0 3 6 9 12 (15) 18	1	15	18		
Hazardous Waste Quantity	0 1 (2) 3 4 5 6 7 8	1	2	8		
Total Waste Characteristics Score			17	26		
5 Targets					3.5	
Ground Water Use	0 1 2 (3)	3	9	9		
Distance to Nearest Well/Population Served	0 4 6 8 10 12 16 18 20 24 30 32 (35) 40	1	35	40		
Total Targets Score			44	49		
6	If line 1 is 45, multiply 1 x 4 x 5 If line 1 is 0, multiply 2 x 3 x 4 x 5		33,600	57,330		
7	Divide line 6 by 57,330 and multiply by 100		S _{gw} = 58.71			

FIGURE 2
GROUND WATER ROUTE WORK SHEET

Carry calculations out to two decimal places

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Surface Water Route Work Sheet							
Rating Factor	Assigned Value (Circle One)		Multi- plier	Score	Max. Score	Ref. (Section)	
1 Observed Release	0	45	1	0	45	4.1	
If observed release is given a value of 45, proceed to line 4 . If observed release is given a value of 0, proceed to line 2 .							
2 Route Characteristics						4.2	
Facility Slope and Intervening Terrain	0	1	2	3	1	3	
1-yr. 24-hr. Rainfall	0	1	2	3	1	3	
Distance to Nearest Surface Water	0	1	2	3	2	6	
Physical State	0	1	2	3	1	3	
Total Route Characteristics Score				NA	15		
3 Containment	0	1	2	3	1	NA	
					3	4.3	
4 Waste Characteristics						4.4	
Toxicity/Persistence	0	3	6	9	12	15	18
Hazardous Waste Quantity	0	1	2	3	4	5	6
Total Waste Characteristics Score				NA	26		
5 Targets						4.5	
Surface Water Use	0	1	2	3	3	9	
Distance to a Sensitive Environment	0	1	2	3	2	6	
Population Served/Distance to Water Intake Downstream	0	4	6	8	10	1	40
				12	16	18	20
				24	30	32	35
Total Targets Score				NA	55		
6 If line 1 is 45, multiply 1 x 4 x 5							
If line 1 is 0, multiply 2 x 3 x 4 x 5							
7 Divide line 6 by 64,350 and multiply by 100				S _{sw} = 0			

FIGURE 7
SURFACE WATER ROUTE WORK SHEET

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Air Route Work Sheet						
Rating Factor	Assigned Value (Circle One)	Multi- plier	Score	Max. Score	Ref. (Section)	
1 Observed Release	0 45	1	0	45	5.1	
Date and Location:						
Sampling Protocol:						
If line 1 is 0, the $S_a = 0$. Enter on line 5 . If line 1 is 45, then proceed to line 2 .						
2 Waste Characteristics					5.2	
Reactivity and Incompatibility	0 1 2 3	1		3		
Toxicity	0 1 2 3	3		9		
Hazardous Waste Quantity	0 1 2 3 4 5 6 7 8	1		8		
Total Waste Characteristics Score			NA	20		
3 Targets					5.3	
Population Within 4-Mile Radius	0 9 12 15 18 21 24 27 30	1		30		
Distance to Sensitive Environment	0 1 2 3	2		6		
Land Use	0 1 2 3	1		3		
Total Targets Score			NA	39		
4 Multiply 1 x 2 x 3			0	35,100		
5 Divide line 4 by 35,100 and multiply by 100			$S_a = 0$			

FIGURE 9
AIR ROUTE WORK SHEET

	s	s ²
Groundwater Route Score (S _{gw})	58.71	3,447.18
Surface Water Route Score (S _{sw})	0	0
Air Route Score (S _a)	0	0
$S_{gw}^2 + S_{sw}^2 + S_a^2$		3,447.18
$\sqrt{S_{gw}^2 + S_{sw}^2 + S_a^2}$		58.71
$\sqrt{S_{gw}^2 + S_{sw}^2 + S_a^2} / 1.73 = S_M =$		33.94

FIGURE 10
WORKSHEET FOR COMPUTING S_M

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that Kysor of Cadillac poses any significant fire or explosion threat

$$\therefore S_{FE} = 0$$

Fire and Explosion Work Sheet						
Rating Factor	Assigned Value (Circle One)		Multi- plier	Score	Max. Score	Ref. (Section)
1 Containment	1	3	1	NA	3	7.1
2 Waste Characteristics						7.2
Direct Evidence	0	3	1		3	
Ignitability	0	1 2 3	1		3	
Reactivity	0	1 2 3	1		3	
Incompatibility	0	1 2 3	1		3	
Hazardous Waste Quantity	0	1 2 3 4 5 6 7 8	1		8	
Total Waste Characteristics Score				NA	20	
3 Targets						7.3
Distance to Nearest Population	0	1 2 3 4 5	1		5	
Distance to Nearest Building	0	1 2 3	1		3	
Distance to Sensitive Environment	0	1 2 3	1		3	
Land Use	0	1 2 3	1		3	
Population Within 2-Mile Radius	0	1 2 3 4 5	1		5	
Buildings Within 2-Mile Radius	0	1 2 3 4 5	1		5	
Total Targets Score				NA	24	
4 Multiply 1 x 2 x 3				0	1,440	
5 Circle line 4 by 1,440 and multiply by 100				SFE = 0		

FIGURE 11
FIRE AND EXPLOSION WORK SHEET

Direct Contact Work Sheet						
Rating Factor	Assigned Value (Circle One)	Multi- plier	Score	Max. Score	Ref. (Section)	
1 Observed Incident	0 45	1	0	45	8.1	
If line 1 is 45, proceed to line 4 If line 1 is 0, proceed to line 2						
2 Accessibility	0 1 2 3	1	3	3	8.2	
3 Containment	0 15	1	0	15	8.3	
4 Waste Characteristics Toxicity	0 1 2 3	5	NA	15	8.4	
5 Targets					8.5	
Population Within a 1-Mile Radius	0 1 2 3 4 5	4		20		
Distance to a Critical Habitat	0 1 2 3	4		12		
Total Targets Score			0	32		
6 If line 1 is 45, multiply 1 x 4 x 5 If line 1 is 0, multiply 2 x 3 x 4 x 5			0	21,600		
7 Divide line 6 by 21,600 and multiply by 100			SDC = 0			

FIGURE 12
DIRECT CONTACT WORK SHEET

DOCUMENTATION RECORDS
FOR
HAZARD RANKING SYSTEM

INSTRUCTIONS: The purpose of these records is to provide a convenient way to prepare an auditable record of the data and documentation used to apply the Hazard Ranking System to a given facility. As briefly as possible summarize the information you used to assign the score for each factor (e.g., "Waste quantity = 4,230 drums plus 800 cubic yards of sludges"). The source of information should be provided for each entry and should be a bibliographic-type reference that will make the document used for a given data point easier to find. Include the location of the document and consider appending a copy of the relevant page(s) for ease in review.

FACILITY NAME: KYSOR OF CADILLAC, INC.

LOCATION: 1100 WRIGHT ST. CADILLAC, MI 49601 WEXFORD CO.

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A/7/10/85
4/11/85

GROUND WATER ROUTE						
1 OBSERVED RELEASE	DEPTH	Toluene	Ethyl Benzene	Trichloroethylene	1,1,1 Trichloroethane	1,1 Dichloroethylene
Contaminants detected (5 maximum):						
UPGRADIENT WELL #17	34'	—	—	—	1.1	—
DOWNGRADIENT WELL #4	28'	140	14	12000	62000	1300
DOWNGRADIENT WELL #14	62'	—	—	2000	760	41
DOWNGRADIENT WELL #15	150'	—	—	530	—	—

[All concentrations in ug/l]

Rationale for attributing the contaminants to the facility:

Taken from Summary of Groundwater Quality Kysor Industrial Corp; Alpha Proj # 2. FTC-H Project No. 81817/163-5 (1). It should be noted that Fishbeck, ^{Phos} Huber & Carr, consulting engineers for Kysor, took these samples in a 3-month period. The samples for the upgradient well #17 were taken on 5/12 & 5/13/83, while well #4, 14, 15 were sampled on 7/26, 7/27+28, & 6/11/83 * respectively. However, the magnitude of the ^{contaminants} concentration

SINCE OBSERVED RELEASE → PROCEED TO SECTION 4. WASTE CHARACTERISTICS. from well #4

2 ROUTE CHARACTERISTICS

Depth to Aquifer of Concern

indicate that the problem indeed originates from Ky. Also, well #15 indicates the depth (150') with which (1) TCE has reached, which is indeed dangerous

Name/description of aquifers(s) of concern: since well #15 borders the Cadillac Municipal well field.

N/A

Depth(s) from the ground surface to the highest seasonal level of the saturated zone [water table(s)] of the aquifer of concern:

N/A

Depth from the ground surface to the lowest point of waste disposal/storage:

N/A

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Net Precipitation

Mean annual or seasonal precipitation (list months for seasonal):

N/A

Mean annual lake or seasonal evaporation (list months for seasonal):

N/A

Net precipitation (subtract the above figures):

N/A

Permeability of Unsaturated Zone

Soil type in unsaturated zone:

N/A

Permeability associated with soil type:

N/A

Physical State

Physical state of substances at time of disposal (or at present time for generated gases):

N/A

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3 CONTAINMENT

Containment

Method(s) of waste or leachate containment evaluated:

N/A

Method with highest score:

N/A

4 WASTE CHARACTERISTICS

Toxicity and Persistence

Compound(s) evaluated:	TOXICITY	PERSISTENCE	MATRIX SCORE
1,1,1 Trichloroethane	2	2	12
Trichloroethylene	2	2	12
1,1 Dichloroethylene	3	2	15
Toluene	2	1	9

(3)

Compound with highest score:

1,1 Dichloroethylene - 15

Hazardous Waste Quantity

Total quantity of hazardous substances at the facility, excluding those with a containment score of 0 (Give a reasonable estimate even if quantity is above maximum): According to SI notes, 28 truckloads (e 35 yd³/load) were excavated from pit. From waste characterization report (Ref #4), the following was determined:

density of sandy soil contaminated w/ organics + misc. residue : 2700 lbs/yd³

total concentration of volatiles, organics, acids, base neutrals : 23,818.17 mg/kg

This results in total quantity = 23.34 yd³ (References # 4, 14)

Basis of estimating and/or computing waste quantity:

$$28 \text{ truckloads} \times \frac{35 \text{ yd}^3}{\text{truck}} \times \frac{2770 \text{ lbs}}{\text{yd}^3} \times \frac{1 \text{ kg}}{2.2 \text{ lbs}} \times \frac{23,818.17 \text{ mg}}{\text{kg}} \times \frac{1 \text{ g}}{1000 \text{ mg}} \times \frac{1 \text{ kg}}{1000 \text{ g}} \times \frac{2.2 \text{ lbs}}{1 \text{ kg}} \times \frac{1 \text{ yd}^3}{2770 \text{ lbs}} = 23.34$$

(Reference # 4, 14)

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owe their geology to a glacial drift. Therefore, the a is in effect an unconfined aquifer ranging from ~6 110 ft. below surface, with entire range subject to contamination. (Ref 8,18). In fact, there is already documentation of private well contamination (15)

5 TARGETS

Ground Water Use

Use(s) of aquifer(s) of concern within a 3-mile radius of the facility:

According to Mr. Don Rennie, Director of Utilities for Cadillac, MI, approximately 92% of the city limit population uses municipal supply. Therefore, about 8% use private wells; some because they prefer to and others because municipal supply is not available. The private wells draw from a shallow aquifer which is linked to the contaminated monitoring wells at Kysor. However, municipal supply comes from deep aquifer (~300 ft) which is most likely separated from shallow aquifer by clay. Also, with very, very few exceptions, population outside Cadillac city limit all use private wells.

Distance to Nearest Well

(Reference 6,7,8,18)

Location of nearest well drawing from aquifer of concern or occupied building not served by a public water supply:

Since the well @ 1033 6th St. is contaminated and is directly downgradient from Kysor, it can be assumed to be in the plume of contamination, + hence the boundary of actual site. Therefore, distance to nearest well = 0 ft.

(15)

Distance to above well or building:

0 ft. (15)

Population Served by Ground Water Wells Within a 3-Mile Radius

Identified water-supply well(s) drawing from aquifer(s) of concern within a 3-mile radius and populations served by each:

The shallow aquifer serves the private wells. Approximately 8% of city population uses private wells. Plus, I counted 956 houses within 3-mile radius from topo m.

∴ Estimated Population = $0.08(10,199) + 956 \text{ houses} \times \frac{3.8 \text{ people}}{\text{house}} = 4448 \text{ people}$

(Reference # 6,9,10,11,12)

Computation of land area irrigated by supply well(s) drawing from aquifer(s) of concern within a 3-mile radius, and conversion to population (1.5 people per acre):

The State of Michigan does not require farmers to apply for usage permits for irrigating farmland with groundwater. Therefore, will assume groundwater not used for irrigation

(13)

Total population served by ground water within a 3-mile radius:

4,448

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1. OBSERVED RELEASE

Contaminants detected in surface water at the facility or downhill from it (5 maximum):

According to Margaret Burns-Guerriero, who was the person in charge of the site inspection, the probability of an overland surface water route for contamination was very unlikely. Therefore, the score

$S_{sw} = 0$. (Reference #19)

Rationale for attributing the contaminants to the facility:

N.A.

2. ROUTE CHARACTERISTICS

Facility Slope and Intervening Terrain

Average slope of facility in percent:

N.A.

Name/description of nearest downslope surface water:

N.A.

Average slope of terrain between facility and above-cited surface water body in percent:

N.A.

Is the facility located either totally or partially in surface water?

N.A.

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Is the facility completely surrounded by areas of higher elevation?

N.A.

1-Year 24-Hour Rainfall in Inches

N.A.

Distance to Nearest Downslope Surface Water

N.A.

Physical State of Waste

N.A.

3 CONTAINMENT

Containment

Method(s) of waste or leachate containment evaluated:

N.A.

Method with highest score:

N.A.

4 WASTE CHARACTERISTICS

Toxicity and Persistence

Compound(s) evaluated

N.A.

Compound with highest score:

N.A.

Hazardous Waste Quantity

Total quantity of hazardous substances at the facility, excluding those with a containment score of 0 (Give a reasonable estimate even if quantity is above maximum):

N.A.

Basis of estimating and/or computing waste quantity:

N.A.

* * *

5 TARGETS

Surface Water Use

Use(s) of surface water within 3 miles downstream of the hazardous substance:

N.A.

Is there tidal influence?

N.A.

Distance to a Sensitive Environment

Distance to 5-acre (minimum) coastal wetland, if 2 miles or less:

N.A.

Distance to 5-acre (minimum) fresh-water wetland, if 1 mile or less:

N.A.

Distance to critical habitat of an endangered species or national wildlife refuge, if 1 mile or less:

N.A.

Population Served by Surface Water

Location(s) of water-supply intake(s) within 3 miles (free-flowing bodies) or 1 mile (static water bodies) downstream of the hazardous substance and population served by each intake:

N.A.

Computation of land area irrigated by above-cited intake(s) and
conversion to population (1.5 people per acre):

N.A.

Total population served:

N.A.

Name/description of nearest of above water bodies:

N.A.

Distance to above-cited intakes, measured in stream miles.

N.A.

AIR ROUTE

1 OBSERVED RELEASE

Contaminants detected:

NONE REPORTED OR MEASURED

$$\therefore S_A = 0$$

Date and location of detection of contaminants

Methods used to detect the contaminants:

Rationale for attributing the contaminants to the site:

* * *

2 WASTE CHARACTERISTICS

Reactivity and Incompatibility

Most reactive compound:

Most incompatible pair of compounds:

Toxicity

Most toxic compound:

Hazardous Waste Quantity

Total quantity of hazardous waste:

Basis of estimating and/or computing waste quantity:

* * *

3 TARGETS

Population Within 4-Mile Radius

Circle radius used, give population, and indicate how determined:

0 to 4 mi

0 to 1 mi

0 to 1/2 mi

0 to 1/4 mi

Distance to a Sensitive Environment

Distance to 5-acre (minimum) coastal wetland, if 2 miles or less:

Distance to 5-acre (minimum) fresh-water wetland, if 1 mile or less:

Distance to critical habitat of an endangered species, if 1 mile or less:

Land Use

Distance to commercial/industrial area, if 1 mile or less:

Distance to national or state park, forest, or wildlife reserve, if 2 miles or less:

Distance to residential area, if 2 miles or less:

Distance to agricultural land in production within past 5 years, if 1 mile or less:

Distance to prime agricultural land in production within past 5 years, if 2 miles or less:

Distance to historic or landmark site (National Historic
National Natural Landmarks) within the view of the site

FIRE AND EXPLOSION

There are no documented threats where either a state or local fire marshal has certified that Kysor of Cadillac poses any significant fire or explosion threat $\therefore S_{FE} = 0$

1 CONTAINMENT

Hazardous substances present:

Type of containment, if applicable:

* * *

2 WASTE CHARACTERISTICS

Direct Evidence

Type of instrument and measurements:

Ignitability

Compound used:

Reactivity

Most reactive compound:

Incompatibility

Most incompatible pair of compounds:

Hazardous Waste Quantity

Total quantity of hazardous substances at the facility:

Basis of estimating and/or computing waste quantity:

* * *

3 TARGETS

Distance to Nearest Population

Distance to Nearest Building

Distance to Sensitive Environment

Distance to wetlands:

Distance to critical habitat:

Distance to

Distance to commercial/industrial area, if 1 mile or less:

Distance to national or state park, forest, or wildlife reserve, if 2 miles or less:

Distance to residential area, if 2 miles or less:

Distance to agricultural land in production within past 5 years, if 1 mile or less:

Distance to prime agricultural land in production within past 5 years, if 2 miles or less:

Is a historic or landmark site (National Register or Historic Places and National Natural Landmarks) within the view of the site?

Population Within 2-Mile Radius

Buildings Within 2-Mile Radius

DIRECT CONTACT

1 OBSERVED INCIDENT

Date, location, and pertinent details of incident:

NONE $\therefore$ LINE 1 = 0

2 ACCESSIBILITY

Describe type of barrier(s):

NO FENCE OR BARRIER (14)

$\therefore$ SCORE = 3

3 CONTAINMENT

Type of containment, if applicable:

Earthen pits ; however pits are no longer there due to clean up
July 22-23, 1981. Wastes are now stored in 55gal. drums in warehouse
until they are transported away

*** $\therefore$ SCORE = 0

Since $S_{DC} = 2 \times 3 \times 4 \times 5$, and $2 = 0$.

4 WASTE CHARACTERISTICS

$\therefore S_{DC} = 0$

Toxicity

Compounds evaluated:

Compound with highest score:

1 TARGETS

Population within one-mile radius

Distance to critical habitat (of endangered species)

HRS DOCUMENTATION LOG SHEET

SITE NAME KYSOR OF CADILLACCITY CADILLACSTATE MIIDENTIFICATION NUMBER MID043631840

REFERENCE NUMBER	DESCRIPTION OF THE REFERENCE
1.	Lab Results & Summary (w/map of site) of monitoring wells at Kysor performed by Alpha Lab; Alpha Project # 21281, 8/24/83, 7/12/83, 6/17/83, 6/9/83, 6/6/83, 5/11/83, 5/20, 5/13
2	Uncontrolled Hazardous Waste Site Ranking System - A User's Manual, Barnett, Chang, Haus, Platt; August 1982 MTR-82W111, pages 20-21, 33
3	Sax, Dangerous Properties of Industrial Materials, copyright 1979 by Litter Educational Publishing Company, 5 th Edition, page 1045.
4	State of Michigan DNR office of Hazardous Waste Management Waste Characterization Report, MID043631840, submitted for Kysor of Cadillac by Chuck Ruotsala, Consulting Engineer, Fishbeck Thompson, Carr & Huber, 6/17/81
5	Addendum to Waste Characterization Report of Kysor of Cadillac, letter written by Chuck Ruotsala, Consulting engineer for Fishbeck Thompson, Carr & Huber, Inc., 8/19/81
6	Telephone conversation between Mr. Don Rennie, Director of Utilities, Cadillac MI and Ron Bock, Chemical Engineer from Ecology and Environment, 12/27/84.
7.	Telephone conversation between Mr. Don Rennie, Director of Utilities, Cadillac, MI and Ron Bock, Chemical Engineer from Ecology and Environment, 12/27/84
8.	Interoffice communication between Claudia Weaver, Environmental Enforcement Division, from Kevin Kinsare, Geologist, Geotechnical Unit, Hydrogeological Section, GWQD, MDNR 12/21/83.

HRS DOCUMENTATION LOG SHEET

SITE NAME Kysor of CadillacCITY CADILLACSTATE MIIDENTIFICATION NUMBER 141004366/1540

REFERENCE NUMBER	DESCRIPTION OF THE REFERENCE
9	Cadillac North Quadrangle, Michigan, 15 minute series topographic 1956
10	Cadillac South Quadrangle, Michigan, 15 minute series topographic 1957
11	Michigan Dept of Public Health, Water Well Records,
12	1980 Census of Population, Number of Inhabitants, Michigan, U.S. Dept of Commerce, Bureau of the Census, Issued January 82.
13	Telephone conversation between Gailen Kilmer, Groundwater Secretary MDNR, and Paul Hess, Chemist, Ecology and Environment, 6/20/84.
14	SI interview at Kysor of Cadillac between Margaret Burns, Hydrogeologist of Ecology and Environment, Inc. and David Crooks, VP and General Counsel of Kysor Industrial Corp, 9/25/84
15	Partial Chemical Analysis of water, collected and reported to Jim Lahti, GWAC, MDNR. Private Well belong to Mike Guest, 1033 6th Ave. Cadillac MI 1/14/81
16	"Endangered Species", U.S. Department of the Interior, Fish and Wildlife Service Great Lakes Region.
17	Telephone conversation between Dr. Sylvia Taylor, MDNR and Ronald Bock, Ecology and Environment, 1/2/85.
18	Telephone Conversation between Mr. Don Rennie, Director of Utilities Cadillac, MI and Ron Bock, Chemical Engineer from E+E, Inc. 10:30 A.M., 4/9/85.
19	Telephone Conversation between Margaret Burns-Guerrero, U.S. EPA & Ron Bock, Chemical Engineer, E+E, Inc., 4/11/85, 9:15 AM, 686-0379